

TABLE I

ACCURACY COMPARISON WITH RECENT ADVANCED CBM BASELINES ON THREE STANDARD BENCHMARKS USING TWO COMMON BACKBONES.

BACKBONE	TYPE	MODEL	#UNITS	ACC.(%) $\uparrow$ (MEAN $\pm$ STD)		
				CUB200	STANFORD CARS	IMAGENET100
ResNet-50	Linear probe	Baseline	2048	79.56 $\pm$ 0.43	85.79 $\pm$ 0.11	86.25 $\pm$ 0.17
	LLM-defined	Label-free CBM	370/456/635	55.26 $\pm$ 2.51	62.99 $\pm$ 0.21	79.76 $\pm$ 0.29
		HybridCBM	2000/-/1040	77.84 $\pm$ 0.23	–	85.20 $\pm$ 0.40
	Auto-discovered	EMC	512	72.47 $\pm$ 0.69	75.82 $\pm$ 0.96	78.76 $\pm$ 0.48
		FLINT	180	74.25 $\pm$ 0.15	79.53 $\pm$ 0.23	80.46 $\pm$ 1.20
		VisCoIN	256	78.86 $\pm$ 0.87	83.34 $\pm$ 0.75	85.09 $\pm$ 0.90
		DCBM	1024	75.82 $\pm$ 0.94	84.38 $\pm$ 0.74	82.55 $\pm$ 0.64
	NFS-CBM (Ours)	128	81.06 $\pm$ 0.12	87.85 $\pm$ 0.05	85.50 $\pm$ 0.07	
ViT-L/14	Linear probe	Baseline	768	86.46 $\pm$ 0.28	87.86 $\pm$ 0.11	89.28 $\pm$ 0.05
	LLM-defined	Label-free CBM	370/456/635	80.43 $\pm$ 0.75	83.66 $\pm$ 0.15	84.04 $\pm$ 0.76
		HybridCBM	2000/-/1040	85.25 $\pm$ 0.68	–	87.60 $\pm$ 0.90
	Auto-discovered	EMC	512	79.32 $\pm$ 0.33	82.53 $\pm$ 0.62	82.87 $\pm$ 0.30
		FLINT	180	82.71 $\pm$ 0.52	82.49 $\pm$ 0.99	84.16 $\pm$ 0.35
		VisCoIN	–	–	–	–
		DCBM	1024	84.45 $\pm$ 0.61	87.33 $\pm$ 0.32	87.10 $\pm$ 0.67
	NFS-CBM (Ours)	128	89.08 $\pm$ 0.27	92.41 $\pm$ 0.17	90.00 $\pm$ 0.34	

Note. Slash-separated values are ordered as CUB200/Cars/ImageNet100. A dash (–) denotes an unavailable result under the corresponding backbone-dataset setting; see Section IV-C.